



Barker
College

--	--	--	--	--	--	--	--

Student Number

2022

YEAR 12
ASSESSMENT TASK 3

Mathematics Extension 2

Staff Involved:

KJL* GDH

35 copies

8:30AM
WEDNESDAY 1 JUNE

TOPICS COVERED

- Vectors / 23
- Integration / 24

General

Instructions:

- Working time – 55 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- Write all answers in the spaces provided
- Show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale

Total marks:

47

Question 1

Marks

If $p = \begin{pmatrix} -3 \\ 1 \\ 5 \end{pmatrix}$ and $q = \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix}$, find the value of:

(i) $p \cdot q$

(ii) $|p - 2q|$

3

Question 2

The coordinates of three points are $P = (3, 5, 2)$, $Q = (-1, 2, 4)$ and $R = (1, 0, 5)$

(i) Using vectors, find the point S such that $PQRS$ is a parallelogram.

3

(ii) Prove that $PQRS$ is in fact a rectangle.

2

Question 3

Find the parametric vector equation of the **interval** AB where $A = (0, 4, 5)$ and $B = (-3, 4, 1)$. **2**

Question 4

Consider two lines, L_1 and L_2 given by the equations below:

$$L_1 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \quad \text{and} \quad L_2 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

Show that L_1 and L_2 are skew lines.

3**Question 5**

Let $A = (-1, 3, 4)$ and $B = (1, 2, 2)$.

(i) Find the angle between the two position vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ in exact form. **2**

(ii) Hence find the exact value of the area of $\triangle OAB$ in simplest surd form. **3**

Question 6

Consider the sphere given by the Cartesian equation $x^2 + y^2 + z^2 + 2x - 4z = 4$.

(i) Show that the vector equation of the sphere is $\left| \vec{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$ **2**

THIS PAGE IS DELIBERATELY LEFT BLANK

(except for this writing)

(ii) Consider the line $\vec{L} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ which intersects with the sphere at two points.

Find the Cartesian coordinates of these two points. **3**

--	--	--	--	--	--	--	--

Student Number

Question 7

Find $\int \frac{1}{\sqrt{7-6x-x^2}} dx$

2

Question 9

Find $\int \frac{2-x}{(x+1)(x^2+1)} dx$

Marks

4

Question 8

Find $\int \sec^4 \theta d\theta$

2

Question 10

(i) If $t = \tan \frac{x}{2}$, show that $dx = \frac{2dt}{1+t^2}$

1

(ii) Hence or otherwise evaluate $\int_0^{\frac{\pi}{2}} \frac{2}{1+\sin x} dx$

3**Question 11**

Using the method of integration by parts, find $\int \sin(\ln x) dx$

3

Question 12

Using an appropriate trigonometric substitution, find $\int \frac{1}{2x\sqrt{16-x^2}} dx$

4**Question 13**

Let $I_n = \int_0^1 \frac{x^{2n}}{x^2+1} dx$ for integers $n \geq 0$

Show that $I_n + I_{n-1} = \frac{1}{2n-1}$ for integers $n \geq 1$

3

Question 14

It is known that for a certain integral, $I = \int_0^{36} [f(x)]^2 dx$.

Find the value of $\int_0^{18} [4f(2x)]^2 dx$, giving your answer in terms of I .

2

Question 1

Marks

If $p = \begin{pmatrix} -3 \\ 1 \\ 5 \end{pmatrix}$ and $q = \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix}$, find the value of:

(i) ~~the~~ $p \cdot q$

$$\begin{aligned} &= (-3)(0) + (1)(2) + (5)(-2) \\ &= 0 + 2 - 10 \\ &= -8 \end{aligned}$$

(ii) $|p - 2q|$

3

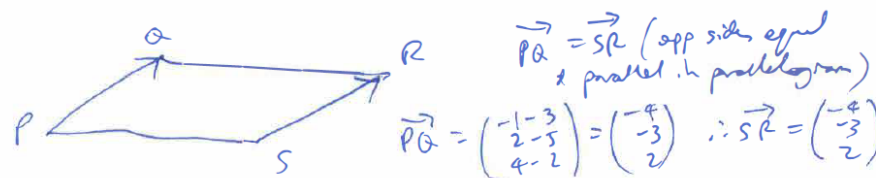
$$\begin{aligned} &= \left| \begin{pmatrix} -3 - 2(0) \\ 1 - 2(2) \\ 5 - 2(-2) \end{pmatrix} \right| = \left| \begin{pmatrix} -3 \\ -3 \\ 9 \end{pmatrix} \right| \\ &= \sqrt{9 + 9 + 81} = \sqrt{99} = 3\sqrt{11} \end{aligned}$$

Question 2

The coordinates of three points are $P = (3, 5, 2)$, $Q = (-1, 2, 4)$ and $R = (1, 0, 5)$

(i) Using vectors, find the point S such that $PQRS$ is a parallelogram.

3



$$\begin{aligned} \vec{OS} &= \vec{OR} + \vec{RS} \\ &= \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix} + \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \\ 3 \end{pmatrix} \quad \therefore S = (5, 3, 3) \end{aligned}$$

(ii) Prove that $PQRS$ is in fact a rectangle.

2

$$\begin{aligned} \vec{QR} &= \begin{pmatrix} 1 - (-1) \\ 0 - 2 \\ 5 - 4 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \\ \vec{QR} \cdot \vec{SR} &= \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ -3 \\ 2 \end{pmatrix} = -8 + 6 + 2 = 0 \end{aligned}$$

$$\therefore \angle QRS = 90^\circ$$

$\therefore PQRS$ is a rectangle (parallelogram with right angle)

END OF PAPER

Question 3

Find the parametric vector equation of the interval AB where $A = (0, 4, 5)$ and $B = (-3, 4, 1)$. 2

$$\vec{AB} = \begin{pmatrix} -3-0 \\ 4-4 \\ 1-5 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ -4 \end{pmatrix}$$

$$\therefore \vec{r} = \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 0 \\ -4 \end{pmatrix} \text{ is the line } AB.$$

For the interval AB , $0 \leq \lambda \leq 1$

$$\therefore \vec{r} = \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 0 \\ -4 \end{pmatrix} \text{ for } 0 \leq \lambda \leq 1$$

Question 4

Consider two lines, L_1 and L_2 given by the equations below:

$$L_1 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \quad \text{and} \quad L_2 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

Show that L_1 and L_2 are skew lines.

Note that direction vectors $\lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$ and $\mu \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ cannot be parallel
as no value of λ or μ satisfies $\lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \mu \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$
[ignoring the trivial case of the zero vector]

Do they intersect?

$$\text{Solve: } \begin{cases} 1 - \lambda = 2 + \mu \\ 3 + \lambda = -1 + 2\mu \\ 2 + \lambda = 3 - \mu \end{cases} \quad \left. \begin{array}{l} \text{Adding vertically, } 3 = 5. \\ \text{Inconsistent equations} \end{array} \right\}$$

$\therefore$ Lines don't intersect

$\therefore$ Lines skew, skew not parallel & don't intersect

or, eqn 2 & 3:

$$1 = -4 + 3\mu$$

$$3\mu = 5 \quad \therefore \mu = \frac{5}{3} \quad \therefore \lambda = -\frac{2}{3}$$

Substitute into eqn 1, does $1 - \frac{2}{3} = 2 + \frac{5}{3}$?

$$\text{Does } \frac{5}{3} = \frac{11}{3} \text{? No.}$$

or, eqn 1 & 2

$$4 = 1 + 3\mu \quad \therefore \mu = 1$$

$$\lambda = -2$$

Substitute into eqn 3, does $2 + 2 = 3 - 1$?

$$3 \quad 0 = 2 \text{? No. Inconsistent equations etc.}$$

Question 5

Let $A = (-1, 3, 4)$ and $B = (1, 2, 2)$.

(i) Find the angle between the two position vectors $\vec{OA}$ and $\vec{OB}$ in exact form. 2

$$\vec{OA} \cdot \vec{OB} = |\vec{OA}| |\vec{OB}| \cos \theta$$

$$-1 + 6 + 8 = \sqrt{1+9+16} \sqrt{1+4+4} \cos \theta$$

$$13 = 3\sqrt{26} \cos \theta$$

$$\therefore \cos \theta = \frac{13}{3\sqrt{26}} = \frac{13\sqrt{26}}{3 \times 26} = \frac{\sqrt{26}}{6}$$

$$\therefore \theta = \cos^{-1}\left(\frac{\sqrt{26}}{6}\right)$$

(ii) Hence find the exact value of the area of $\triangle OAB$ in simplest surd form. 3

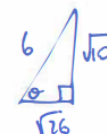
$$\frac{1}{2} |\vec{OA}| |\vec{OB}| \sin \angle AOB$$

$$= \frac{1}{2} \sqrt{26} \cdot 3 \cdot \frac{\sqrt{10}}{6}$$

$$= \frac{1}{4} \sqrt{260}$$

$$= \frac{1}{4} \sqrt{4 \times 65}$$

$$= \frac{\sqrt{65}}{2} \text{ u}^2$$



Question 6

Consider the sphere given by the Cartesian equation $x^2 + y^2 + z^2 + 2x - 4z = 4$.

(i) Show that the vector equation of the sphere is $\left| \underline{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$

2

$$x^2 + 2x + 1 + y^2 + z^2 - 4z + 4 = 4 + 1 + 4$$

$$(x+1)^2 + y^2 + (z-2)^2 = 9$$

$\therefore$ sphere has centre $(-1, 0, 2)$, radius 3

$$\text{i.e. } \left| \underline{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$$

(ii) Consider the line $\underline{L} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ which intersects with the sphere at two points.

Find the Cartesian coordinates of these two points.

3

$$\left| \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} \lambda \\ \lambda \\ 2\lambda \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$$

$$\left| \begin{pmatrix} \lambda+1 \\ \lambda+1 \\ 2\lambda-1 \end{pmatrix} \right| = 3$$

$$\therefore \sqrt{(\lambda+1)^2 + (\lambda+1)^2 + (2\lambda-1)^2} = 3$$

$$\begin{aligned} 6\lambda^2 + 3 &= 9 \\ 6\lambda^2 &= 6 \\ \lambda^2 &= 1 \\ \lambda &= \pm 1 \end{aligned}$$

$$\therefore \text{Points are } \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\text{and } \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

Check that both satisfy

$$5 \quad \left| \underline{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3 \quad \checkmark \text{ Both}$$

Question 7

Find $\int \frac{1}{\sqrt{7-6x-x^2}} dx$

2

$$= \int \frac{dx}{\sqrt{-(x^2+6x-7)}} = \int \frac{dx}{\sqrt{-(x+3)^2-16}} = \int \frac{dx}{\sqrt{16-(x+3)^2}}$$

$$= \sin^{-1}\left(\frac{x+3}{4}\right) + C$$

Question 8

Find $\int \sec^4 \theta d\theta$

2

$$\begin{aligned} &= \int \sec^2 \theta \sec^2 \theta d\theta \\ &= \int \sec^2 \theta (1 + \tan^2 \theta) d\theta \\ &= \int \sec^2 \theta + \sec^2 \theta \tan^2 \theta d\theta \\ &= \tan \theta + \frac{\tan^3 \theta}{3} + C \end{aligned}$$

Question 9

Find $\int \frac{2-x}{(x+1)(x^2+1)} dx$

$$\text{let } \frac{A}{x+1} + \frac{Bx+C}{x^2+1} = \frac{2-x}{(x+1)(x^2+1)}$$

$$\therefore A(x^2+1) + (Bx+C)(x+1) = 2-x$$

$$\text{let } x = -1 \quad \therefore 2A = 3 \quad \therefore A = \frac{3}{2}$$

$$\text{Coeff } x^2: A+B=0 \quad \therefore B = -\frac{3}{2}$$

$$\text{Constant: } A+C=2 \quad \therefore C = \frac{1}{2}$$

$$\therefore \int \frac{\frac{3}{2}}{x+1} + \frac{\frac{1}{2} - \frac{3}{2}x}{x^2+1} dx$$

$$= \frac{3}{2} \int \frac{1}{x+1} dx + \frac{1}{2} \int \frac{1}{x^2+1} dx - \frac{3}{2} \int \frac{x}{x^2+1} dx$$

$$= \frac{3}{2} \ln|x+1| + \frac{1}{2} \tan^{-1}x - \frac{3}{4} \ln|x^2+1| + C$$

Marks

4

Question 10

(i) If $t = \tan \frac{x}{2}$, show that $dx = \frac{2dt}{1+t^2}$

$$\tan^{-1}t = \frac{x}{2}$$

$$\therefore x = 2 \tan^{-1}t$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

$$\therefore dx = \frac{2dt}{1+t^2}$$

$$\begin{aligned} \text{or } \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} \\ &= \frac{1}{2} (1 + \tan^2 \frac{x}{2}) \\ \frac{dt}{dx} &= \frac{1}{2} (1 + t^2) \\ \therefore 2dt &= dx(1+t^2) \\ \therefore dx &= \frac{2dt}{1+t^2} \end{aligned}$$

1

(ii) Hence or otherwise evaluate $\int_0^{\frac{\pi}{2}} \frac{2}{1+\sin x} dx$

$$\text{let } t = \tan \frac{x}{2}$$

$$x=0: t=0$$

$$x=\frac{\pi}{2}: t = \tan \frac{\pi}{4} = 1$$



$$\therefore \int_0^1 \frac{2}{1 + \frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$= 4 \int_0^1 \frac{dt}{1+t^2+2t}$$

$$= 4 \int_0^1 \frac{dt}{(t+1)^2}$$

$$= 4 \int_0^1 (t+1)^{-2} dt$$

$$= 4 \left[\frac{(t+1)^{-1}}{-1} \right]_0^1$$

$$= -4 \left[\frac{1}{1+t} \right]_0^1$$

$$= -4 \left[\frac{1}{2} - 1 \right]$$

$$= 2$$

3

Question 11

Using the method of integration by parts, find $\int \sin(\ln x) dx$

$$\text{let } u' = 1, v = \sin(\ln x)$$

$$\therefore u = x, v' = \frac{\cos(\ln x)}{x}$$

$$\therefore I = \int \sin(\ln x) dx = x \sin(\ln x) - \int x \frac{\cos(\ln x)}{x} dx$$

$$I = x \sin(\ln x) - \int \cos(\ln x) dx$$

$$\text{let } u' = 1, v = \cos(\ln x)$$

$$\therefore u = x, v' = \frac{-\sin(\ln x)}{x}$$

$$\therefore I = x \sin(\ln x) - \left[x \cos(\ln x) - \int x \left(\frac{-\sin(\ln x)}{x} \right) dx \right]$$

$$I = x \sin(\ln x) - x \cos(\ln x) + \int \sin(\ln x) dx$$

$$\therefore 2I = x (\sin(\ln x) - \cos(\ln x))$$

$$\therefore I = \frac{x (\sin(\ln x) - \cos(\ln x))}{2}$$

Question 12

Using an appropriate trigonometric substitution, find $\int \frac{1}{2x\sqrt{16-x^2}} dx$

4

$$\text{let } x = 4 \cos \theta \quad \therefore \frac{dx}{d\theta} = -4 \sin \theta \quad \therefore dx = -4 \sin \theta d\theta$$

$$\therefore = \int \frac{-4 \sin \theta d\theta}{2(4 \cos \theta) \sqrt{16 - 16 \cos^2 \theta}}$$

$$\text{or let } x = 4 \sin \theta$$

$$= -\frac{1}{2} \int \frac{\sin \theta d\theta}{\cos \theta \sqrt{1 - \cos^2 \theta}}$$

$$= -\frac{1}{8} \int \frac{\sin \theta d\theta}{\cos \theta \sin \theta}$$

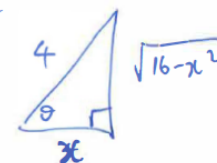
$$= -\frac{1}{8} \int \sec \theta d\theta$$

$$= -\frac{1}{8} \int \frac{\sec \theta (\sec \theta + \tan \theta) d\theta}{\sec \theta + \tan \theta}$$

$$= -\frac{1}{8} \ln |\sec \theta + \tan \theta|$$

$$= -\frac{1}{8} \ln \left| \frac{4}{x} + \frac{\sqrt{16-x^2}}{x} \right|$$

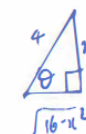
$$= -\frac{1}{8} \ln \left| \frac{4 + \sqrt{16-x^2}}{x} \right| + C$$



$$= -\frac{1}{8} \int \frac{\csc \theta (\cot \theta + \csc \theta) d\theta}{\cot \theta + \csc \theta}$$

$$= -\frac{1}{8} \ln |\cot \theta + \csc \theta|$$

$$\text{or } \frac{1}{8} \ln |\csc \theta - \cot \theta|$$



Question 13

Let $I_n = \int_0^1 \frac{x^{2n}}{x^2+1} dx$ for integers $n \geq 0$

Show that $I_n + I_{n-1} = \frac{1}{2n-1}$ for integers $n \geq 1$

3

$$\begin{aligned} I_n &= \int_0^1 \frac{x^{2n}}{x^2+1} dx \\ &= \int_0^1 \frac{(x^2+1)x^{2n-2}}{x^2+1} dx \\ &= \int_0^1 \frac{x^{2n-2}}{1} dx - \int_0^1 \frac{x^{2n-2}}{x^2+1} dx \\ &= \int_0^1 x^{2n-2} dx - \int_0^1 \frac{x^{2(n-1)}}{x^2+1} dx \\ &= \left[\frac{x^{2n-1}}{2n-1} \right]_0^1 - I_{n-1} \end{aligned}$$

$$I_n = \left(\frac{1}{2n-1} - 0 \right) - I_{n-1}$$

$$I_n + I_{n-1} = \frac{1}{2n-1}$$

OR

$$\begin{aligned} I_n + I_{n-1} &= \int_0^1 \frac{x^{2n}}{x^2+1} dx + \int_0^1 \frac{x^{2n-2}}{x^2+1} dx \\ &= \int_0^1 \frac{x^{2n} + x^{2n-2}}{x^2+1} dx \\ &= \int_0^1 \frac{x^{2n-2}(x^2+1)}{x^2+1} dx \\ &= \int_0^1 x^{2n-2} dx \\ &= \left[\frac{x^{2n-1}}{2n-1} \right]_0^1 = \frac{1}{2n-1} \end{aligned}$$

Question 14

It is known that for a certain integral, $I = \int_0^{36} [f(x)]^2 dx$.

Find the value of $\int_0^{18} [4f(2x)]^2 dx$, giving your answer in terms of I .

2

$$= \int_0^{18} 16 [f(2x)]^2 dx$$

$$= 16 \int_0^{18} [f(2x)]^2 dx$$

$$\begin{aligned} \text{Let } u &= 2x \\ \frac{du}{dx} &= 2 \\ \therefore dx &= \frac{du}{2} \end{aligned}$$

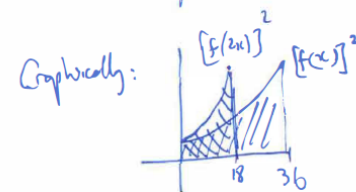
$$\begin{aligned} x=0 &\therefore u=0 \\ x=18 &\therefore u=36 \end{aligned}$$

$$\therefore = 16 \int_0^{36} [f(u)]^2 \frac{du}{2}$$

$$= 8 \int_0^{36} [f(u)]^2 du$$

$$= 8 \int_0^{36} [f(x)]^2 dx \quad \text{with change of variable}$$

$$= 8I$$



$$\text{In a sense, } \int_0^{36} [f(x)]^2 dx = 2 \int_0^{18} [f(2x)]^2 dx$$

since horizontally everything is doubled

$$\therefore 8 \int_0^{36} [f(x)]^2 dx = 16 \int_0^{18} [f(2x)]^2 dx$$

$$\therefore 8I = \int_0^{18} [4f(2x)]^2 dx$$

END OF PAPER